

Project 5: Discretionary Access Control

- Due Mar 15 by 11:59pm
- Points 50
- Submitting an external tool
- Available Mar 9 at 12am - Mar 15 at 11:59pm

This assignment was locked Mar 15 at 11:59pm.

Step 1: Accessing the Server

Log in using SSH:

```
ssh <username>@23.92.18.190
```

Retrieving Your Credentials

1. Visit [this link](https://github.com/cy2550repo/cy2550/blob/dc3fa37470bb4801be38d03018330fc35998594f/Project5/email)  (<https://github.com/cy2550repo/cy2550/blob/dc3fa37470bb4801be38d03018330fc35998594f/Project5/email>) to obtain your login credentials.
2. Search for your email address in the file. The long string following `<your_email>:` is your password.
3. Your username is the part of your email **before** the `@` symbol. For example, if your email is `hs.wang@northeastern.edu`, then your username would be `hs.wang`.

Important: Use only your own account. Flags are unique to each student. Using another student's flag will result in a **zero** for the assignment.

Step 2: Finding the Flags

Within your **home directory**, you will find **ten level directories**. Your objective is to locate a flag in each directory.

- Submit both **the flags** and **screenshots** of your method for finding them. **We expect the screenshots to detail the entire process for finding the flag.**
- Flags follow the format `CY2550_FLAG_LEVELn{<32 characters>}`, where **n** represents the level number.

Hints

Level 1: Basic File Access

- Why can't you read the file?

Level 2: Permission Issues

- A more complex access control challenge.
- **Relevant topics:** Permission bits.
- **Relevant commands:** `./<executable name>`.

Level 3: Encoded String

- The flag appears distorted—what's wrong?
- **Relevant topics:** Alphabetic substitution ciphers, binary-to-text encoding schemes.
- **Relevant commands:** `tr`.
- **Useful tool:** [CyberChef](https://cyberchef.org/)  (<https://cyberchef.org/>).

Level 4: Hidden Image Data

- How might information be inserted into an image?
- **Relevant topics:** Image metadata.
- **Relevant commands:** `exiftool`.

Level 5: Anomalous Image

- There's something odd about this image.
- **Relevant commands:** `binwalk`, `zip`, `unzip`.

Level 6: Privileged Commands

- Which commands can this user run with elevated privileges, and what can those commands reveal about a file without opening it normally?
- **Relevant topics:** Privilege escalation, `sudo` permissions, printable strings in files.
- **Relevant commands:** `sudo`, `sudo -l`, `strings`.

Level 7: Command Actions

- You are allowed to run a command with elevated privileges, but running it directly is not enough. How might you use one of that command's actions to execute another command that reads the flag?
- **Relevant topics:** Command execution, program actions, privilege escalation through allowed commands.
- **Relevant commands:** `sudo`, `find`, `cat`.
- **Special syntax:** `-exec <command> {} \;`.

- Look at the ACTIONS section of the `find` man page. Some actions allow find to execute other commands.

Level 8: Hidden File Discovery

- The flag exists inside the directory, but it is not immediately visible. Instead, the flag is inside a .txt file that begins with 4 digit number. Can you search through the possible filenames efficiently?
- **Relevant topics:** File enumeration, filename patterns, simple shell scripting.
- **Relevant commands:** `cat`, `ls`.

Level 9: Program Interaction

- How might the program's built-in features let you open or read a file that is otherwise restricted?
- **Relevant topics:** SUID programs, file access through privileged binaries, built-in editor functionality.
- **Relevant commands:** `./<executable name>`.

Level 10: Executable Modification

- One program runs with elevated privileges and calls another file that you are able to modify. How could you change that second file so that running the privileged program leaves you able to read the flag directly?
- **Relevant topics:** SUID behavior, script modification, privileged shell execution.
- **Relevant commands:** `./<executable name>`, `nano`, `cat`.

Step 3: Submitting Your Work

Autograded Submission — 50 points

1. Create a directory named `project5` in your GitHub repository.
2. Inside `project5`, create a file named `email.txt` that contains your full Northeastern email address.
3. Inside `project5`, create a file named `flags.txt` that contains your ten flags, each on a newline.
4. Add, commit, and push your changes.
5. Submit your GitHub repository to “Project 5: Discretionary Access Control” on Gradescope.
6. **Check the autograder and resubmit until you see the score you are hoping for!**

Screenshot Submission (Gradescope) — 100 points

- Submit all screenshots to “Project 5: Discretionary Access Control - PDF/Images Submission” on Gradescope.
- Screenshots should clearly show each step you took to retrieve each flag.

Additional Notes

- Use Discord for questions, but do not share answers or post solutions.